

COMPETITIVE INTELLIGENCE REPORT

DARPin & Novel Scaffold Therapeutics

Strategic Positioning Analysis for Molecular Partners AG

Prepared Exclusively for

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Disclaimer & Methodology

Data Sources: All data points sourced from public registries accessed 27 May 2026: ClinicalTrials.gov, EMA Human Medicines Register, PubMed.

Intended Use: Internal strategic use by Molecular Partners AG management.

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1. Executive Summary

Molecular Partners AG (SIX: MOLN) is a Swiss clinical-stage biopharmaceutical company developing DARPin® (Designed Ankyrin Repeat Protein) therapeutics — a novel class of biologic drugs that combine the specificity of antibodies with the stability and manufacturability of small proteins. This report analyzes the DARPin and novel scaffold landscape and Molecular Partners' position within it.

1.1 Key Findings

1. Two Novel Mechanisms in Ph1/2: MP0533 (tri-specific DARPin CD3-engager targeting CD33, CD123, CD70 in AML) and MP0712 (²¹²Pb radiopharmaceutical DARPin targeting DLL3 in SCLC/NETs) represent two distinct, first-in-class approaches. Both are RECRUITING.

2. MP0533 is a Potential Paradigm Shift in AML: The tri-specific design targeting three AML antigens (CD33, CD123, CD70) simultaneously is a novel approach to overcome antigen escape — a major limitation of existing CD33 and CD123 bispecifics. Preclinical data (Cancer Immunol Res 2024, PMID 38683145) demonstrated selective killing of AML leukemic stem cells.

3. MP0712 Enters the Radiopharmaceutical Race: The ²¹²Pb-labeled DARPin targeting DLL3 positions Molecular Partners in the rapidly growing radiopharmaceutical space. Partnering with Orano Med provides radiochemistry expertise. DLL3 is a validated target in SCLC (tarlatamab, Amgen, Ph3).

4. DARPin Platform — Proven but Underleveraged: The DARPin platform has generated clinical data across oncology (MP0250, MP0274) and COVID-19 (ensovibep), but no drug has reached approval. The shift toward wholly owned assets (MP0533, MP0712) represents a strategic pivot from the previous partnered model.

5. Financial Position — Public Company with Focused Pipeline: As a publicly traded company (SIX: MOLN), Molecular Partners has transparency on cash runway. With ~CHF 100-150M market cap and 3 active trials, strategic capital allocation is critical — particularly with MP0317 (biliary tract) and MP0533 (AML) both requiring Phase 2 funding.

Source: ClinicalTrials.gov (NCT05673057, NCT07278479, NCT07036380); PubMed (PMID 38683145, 42048463); SIX Swiss Exchange filings

2. Company & Pipeline Overview

2.1 Corporate Profile

Parameter	Value	Source
Company Name	Molecular Partners AG	Corporate
Ticker	MOLN (SIX Swiss Exchange)	SIX
Headquarters	Schlieren/Zürich, Switzerland	Corporate
CEO	Patrick Amstutz, PhD (co-founder)	Corporate
Market Cap	~CHF 100-150M	SIX
Platform	DARPin® — Designed Ankyrin Repeat Proteins	Corporate / PubMed
Lead Assets	MP0533 (AML), MP0712 (SCLC/NETs), MP0317 (BTC)	ClinicalTrials.gov
Stage	Three wholly owned assets in Ph1/2	ClinicalTrials.gov

2.2 Clinical Pipeline

Asset	Target	Modality	Indication	Phase	Status	Enrollment
MP0533	CD33 × CD123 × CD70	Tri-specific DARPin CD3-engager	AML/MDS	Ph1/2	RECRUITING	249
MP0712	DLL3	²¹² Pb-DARPin radiopharmaceutical	SCLC, LCNEC, epNEC	Ph1/2	RECRUITING	138
MP0317	FAP × CD40	Tri-specific DARPin, tumor-localized CD40 agonist	Biliary tract cancer	Ph1/2	RECRUITING	—

Source: ClinicalTrials.gov (NCT05673057, NCT07278479, NCT07036380), accessed 2026-05-27

2.3 The DARPin Advantage

Parameter	DARPin	Conventional mAb	Bispecific mAb
Size	~14-18 kDa (single domain)	~150 kDa	~150-200 kDa
Tumor Penetration	High (deeper tissue diffusion)	Moderate	Moderate

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Multispecificity	Built-in (single molecule)	No	Yes (engineered)
Stability	Very high (Tm >70°C)	High	High
Manufacturing	E. coli expression (low COGS)	CHO cells (high COGS)	CHO cells (high COGS)
Immunogenicity	Low (fully human design)	Variable	Variable
Flexibility	Modular — swap domains for new targets	Requires new antibody	Requires new engineering

Source: Molecular Partners corporate materials; PubMed (DARPin platform publications)

3. Therapeutic Landscape & Competition

3.1 AML — The MP0533 Opportunity

Acute myeloid leukemia (AML) remains a high-unmet-need indication with limited treatment options beyond chemotherapy and targeted agents for select mutations. Key competitive dynamics:

- **CD33-targeted:** Gemtuzumab ozogamicin (Mylotarg, Pfizer) — the only approved CD33 ADC; limited by toxicity
- **CD123-targeted:** Tagraxofusp (Elzonris) — approved for BPDCN, being explored in AML
- **Bispecific T-cell engagers:** Multiple CD33xCD3 and CD123xCD3 programs in Ph1/2 (Amgen, MacroGenics, etc.)
- **MP0533 differentiation:** First tri-specific approach targeting three antigens simultaneously — reduces antigen escape risk, which is the primary failure mode for single-target bispecifics in AML

3.2 DLL3-Targeting — The MP0712 Opportunity

DLL3 (delta-like ligand 3) is an emerging target in SCLC and neuroendocrine tumors:

- **Tarlatamab (Amgen, Imdelltra):** DLL3xCD3 bispecific T-cell engager — approved in US (2024) for 2L+ SCLC. Establishes DLL3 as a clinically validated target
- **MP0712 differentiation:** ²¹²Pb alpha-emitting radiopharmaceutical — different mechanism (radiation, not T-cell) — avoids CRS/ICANS toxicity of T-cell engagers
- **Orano Med partnership:** Provides access to ²¹²Pb supply chain and radiopharmaceutical expertise

3.3 Novel Scaffold Competition

Platform	Company	Scaffold Type	Stage
DARPin®	Molecular Partners	Ankyrin repeat protein	Ph1-2 (oncology)
Affimer	Avacta Group	Cystatin scaffold	Preclinical-Ph1
Adnectin	BMS (formerly Adnexus)	Fibronectin type III domain	Discontinued
Anticalin	Pieris Pharmaceuticals	Lipocalin scaffold	Ph1-2 (oncology)
Kunitz domain	Various	Protease inhibitor domain	Ph2-3 (non-oncology)
Centyrin	Janssen	Consensus tenascin scaffold	Ph1

Source: Company filings; ClinicalTrials.gov; industry literature, accessed 2026-05-27

4. Key Catalysts & Strategic Positioning

4.1 Near-Term Catalysts

Event	Timeline	Impact
MP0533 Ph1 dose-escalation data update	2026-2027	First clinical data for tri-specific DARPin T-cell engager in AML
MP0712 Ph1 initial safety data	2026-2027	First-in-human data for ²¹² Pb-DARPin radiopharmaceutical
MP0317 Ph2 biliary tract cancer data	2027	FAP-targeted CD40 agonist — novel IO mechanism
MP0533 Ph2 dose optimization	2028	ORR data in AML (n=249 trial)

4.2 Strategic Recommendations

1. MP0533 is the Portfolio Anchor. The tri-specific, multi-antigen approach in AML addresses a clear unmet need (antigen escape). Strong preclinical data (Cancer Immunol Res 2024) de-risks the mechanism. Resources should prioritize generating clinical proof-of-concept.

2. MP0712 Taps into the Radiopharmaceutical Boom. The radiopharmaceutical space is attracting significant pharma interest (Novartis/AAA, Eli Lilly/POINT Biopharma, BMS/RayzeBio). DLL3 is validated (tarlatamab), and alpha-emitters are the next frontier. The Orano Med partnership is strategically sound.

3. Platform Story Needs a Win. Despite strong science, no DARPin has reached approval. The current wholly owned pipeline is Molecular Partners' best chance to demonstrate that DARPin is a viable drug modality, not just a research tool.

4. Cash Management is Critical. With ~CHF 100-150M market cap and 3 active Ph1/2 trials, financing the pipeline to meaningful data readouts requires disciplined capital allocation. BD partnerships for non-core assets (MP0317?) could extend runway.

5. DARPin Versatility is an Undervalued Asset. The platform's modularity (swap domains = new program) is not reflected in the current market cap. A positive clinical readout for any single asset would likely re-rate the entire platform.

5. Key Takeaways for Patrick Amstutz

1. MP0533 is Molecular Partners' Most Important Program. The tri-specific design is genuinely differentiated from the dozens of CD3-engagers in development. Clinical proof that a tri-specific DARPin T-cell engager works in AML would validate both the asset and the platform.

2. Radiopharmaceutical Opportunity has a Narrow Window. Pharma interest in radiopharmaceuticals is at an all-time high. Early data from MP0712 that demonstrates tumor targeting and acceptable dosimetry in SCLC/neuroendocrine patients would position Molecular Partners for a high-value partnership.

3. Pipeline Breadth is a Double-Edged Sword. Three wholly owned assets across three different modalities (T-cell engager, radiopharmaceutical, tumor-localized IO) provides optionality but dilutes resources. Prioritization of one or two programs to definitive clinical data may be necessary.

4. The DARPin Platform Deserves a BD Partnering Strategy. Other novel scaffolds (Anticalin/ Pieris, Affimer/Avacta) have successfully partnered with pharma. Molecular Partners should leverage the platform's advantages (multispecificity, stability, manufacturing) in proactive BD discussions — particularly for targets where DARPin's small size and tissue penetration provide clear advantages over antibodies.